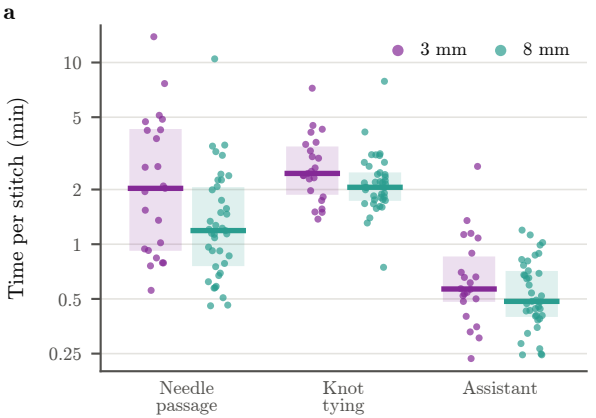
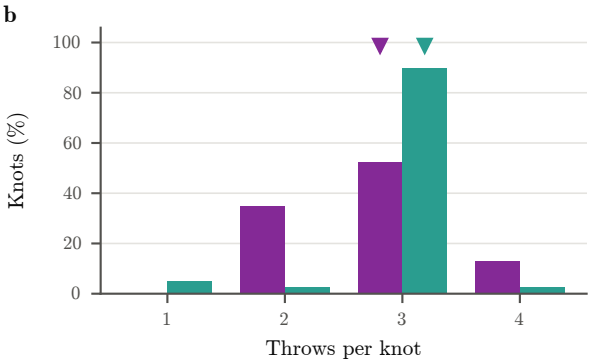
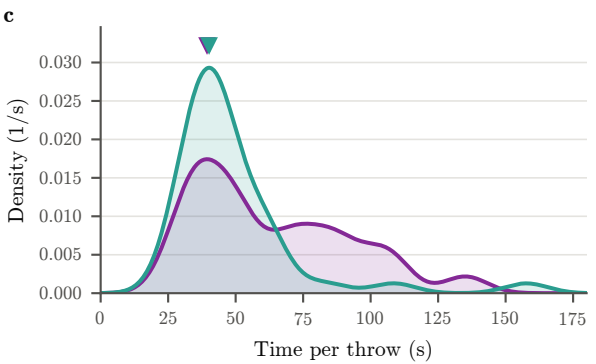

a
3 mm
8 mm
Time per stitch (min)
10
5
2
1
0.5
0.25
Needle passage
Knot tying
Assistant


b
Knots (%)
100
80
60
40
20
0
1
2
3
4
Throws per knot


c
Density (1/s)
0.030
0.025
0.020
0.015
0.010
0.005
0.000
0
25
50
75
100
125
150
175
Time per throw (s)